

Harshit Agarwal

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PROFESSIONAL SUMMARY

Engineer focused on building and deploying real-world AI products. Experienced across full-stack and data workflows using Next.js, Python, PostgreSQL, PGVector, Pipecat, and n8n, with hands-on exposure to microservices, vector search, and cloud-based delivery. Passionate about practical AI engineering—integrating LLM-powered features, scalable architectures, and automated workflows to build products that solve real business problems at scale.

SKILLS

Languages: Python, JavaScript, SQL, CLI

Frontend: Next.js, React

Backend: REST APIs, SQLAlchemy

Databases: MySQL, PostgreSQL

AI & Systems: Vector Search (PGVector), Voice AI (Pipecat), Document AI & OCR, Workflow Automation (n8n), Microservices

Tools & Platforms: Git, GitHub, Postman, VS Code, Google Cloud Run

EXPERIENCE

Vaanii Global Technologies LLP

Jun 2026 – Present

Founding Software Engineer

Remote, India

- Own the end-to-end backend architecture for an AI-powered customer engagement platform, built independently from the ground up.
- Built a new **Voice AI backend** integrating LLM, Speech-to-Text, and Text-to-Speech providers into a unified conversation pipeline.
- Built a production **OCR-based document verification pipeline** for government documents, cross-verifying a submitted document against 10+ supporting documents at the field level.
- Achieved **99.9% text extraction accuracy** with table structure preserved, processing a complete 20-page verification set end-to-end in **33–40 seconds** across extraction, verification, and response.
- Designed CRM and campaign-management features (lead upload, status tracking, outbound voice campaigns) backed by asynchronous processing pipelines on **FastAPI**, **SQLAlchemy**, and **PostgreSQL**, deployed on **Google Cloud**.

ConversAI Labs

Jun 2025 – Jun 2026

Software Engineer (prev. Full Stack Intern)

Remote, India

- Led development of the **Pipecat-based Voice AI backend**, designed to scale to **900+ calls/hour**; handled **480+ calls/hour** in production under telephony provider concurrency limits.
- Built backend services with **FastAPI** and **SQLAlchemy**, and automated post-call workflows (WhatsApp messaging, form submission, CRM logging) end-to-end in n8n, removing manual lead processing.
- Reduced conversation latency and mid-call drop-offs by optimizing turn-taking logic in production.
- Built PGVector-based retrieval across 100+ transcripts to improve AI response accuracy.

PROJECTS

Plant Leaf Disease Detection | *Python, PyTorch, Streamlit* | [Link](#)

- Built a **PyTorch CNN** model using pre-trained **ResNet50** to classify plant leaf diseases, trained on **87,000+ images** across multiple disease categories.
- Developed an interactive **Streamlit** web app enabling real-time image upload and instant disease prediction.
- Implemented image preprocessing pipelines (resizing, normalization) to standardize inputs and improve model consistency.

Uber Clone Application | *React Native, JavaScript, APIs* | [Link](#)

- Developed a full-featured **Uber Clone** in React Native with real-time ride estimation and **location tracking** via third-party API integrations.
- Implemented secure **user authentication and authorization** flows to protect application access and user data.

EDUCATION

Krishna Engineering College

B.Tech in Computer Science

Aug 2022 – May 2025

Ghaziabad, Uttar Pradesh